



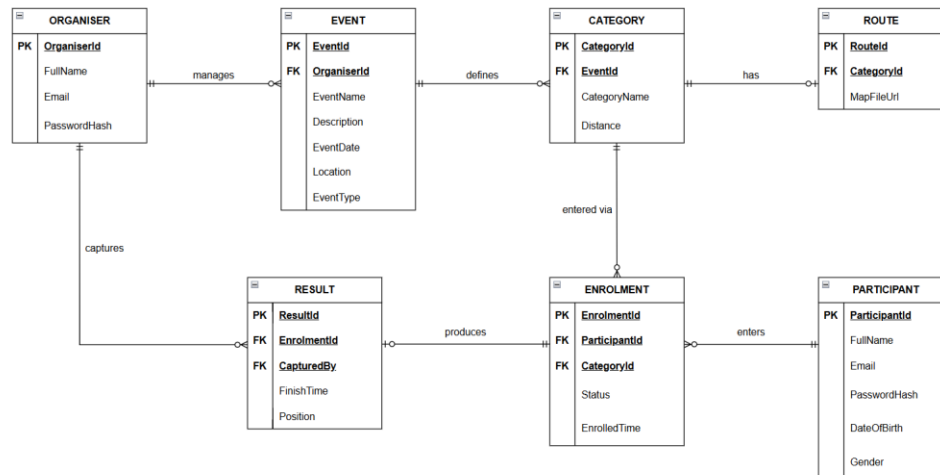
POE PART 1

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Assessment type: POE (part 1)

Section A: ERD



Section B: API Endpoint Plan

This plan covers Authentication, User Profile, Events, Categories, Event Enrolments and Results, plus Route endpoints.

Organiser and Participant are separate tables (no shared Users table with a Role column). Role is therefore managed by *which endpoint/table the account is created in*, registration is split into two endpoints below, and login checks both tables to determine the role of the returned session.

Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register/organiser	Registers a new Organiser account.	None (public)	{ fullName, email, password }	201 Created – organiser profile (no password) / 400 Bad Request – validation errors / 409 Conflict – email already registered

POST	/api/auth/register/participant	Registers a new Participant account.	None (public)	{ fullName, email, password, dateOfBirth, gender }	201 Created – participant profile (no password) / 400 Bad Request / 409 Conflict – email already registered
POST	/api/auth/login	Authenticates a user against both tables and starts a session with the correct role.	None (public)	{ email, password }	200 OK – session started, returns id, name, role / 401 Unauthorized – invalid credentials
POST	/api/auth/logout	Ends the current user's session.	Any (logged in)	None	200 OK – session ended / 401 Unauthorized

User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Retrieves the logged-in user's own profile.	Any (logged in)	None	200 OK – profile object / 401 Unauthorized
PUT	/api/users/profile	Updates the logged-in user's own profile.	Any (logged in)	{ fullName, email }	200 OK – updated profile / 400 Bad Request / 401 Unauthorized

Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all events.	None (public)	None	200 OK – list of events
GET	/api/events/{id}	Retrieves a single event's details.	None (public)	None	200 OK – event object / 404 Not Found
POST	/api/events	Creates a new event owned by the logged-in Organiser.	Organiser	{ eventName, description, eventDate, location, eventType }	201 Created – event object / 400 Bad Request / 401 Unauthorized / 403 Forbidden
PUT	/api/events/{id}	Updates an event owned by the logged-in Organiser.	Organiser	{ eventName, description, eventDate, location, eventType }	200 OK – updated event / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
DELETE	/api/events/{id}	Deletes an event owned by the logged-in Organiser.	Organiser	None	200 OK – confirmation / 401 Unauthorized / 403 Forbidden / 404 Not Found

Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists all categories for an event.	None (public)	None	200 OK – list of categories / 404 Not Found

POST	/api/events/{eventId}/categories	Creates a category under an event owned by the logged-in Organiser.	Organiser	{ categoryName, distance }	201 Created – category object / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
PUT	/api/categories/{id}	Updates a category belonging to the logged-in Organiser's event.	Organiser	{ categoryName, distance }	200 OK – updated category / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
DELETE	/api/categories/{id}	Deletes a category.	Organiser	None	200 OK – confirmation / 401 Unauthorized / 403 Forbidden / 404 Not Found

Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments	Enrols the logged-in Participant into a category.	Participant	{ categoryId }	201 Created – enrolment object / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found / 409 Conflict – already enrolled
GET	/api/enrolments/my	Lists all enrolments belonging to	Participant	None	200 OK – list of enrolments / 401 Unauthorized

		the logged-in Participant.			
GET	/api/events/{eventId}/enrolments	Lists all enrolments for an event owned by the logged-in Organiser.	Organiser	None	200 OK – list of enrolments / 401 Unauthorized / 403 Forbidden / 404 Not Found

Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Captures a finish time and position for a Participant's enrolment.	Organiser	{ enrolmentId, finishTime, position }	201 Created – result object / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
GET	/api/results/my	Lists all results belonging to the logged-in Participant.	Participant	None	200 OK – list of results / 401 Unauthorized
GET	/api/results/{id}	Retrieves a single result (owning Participant or the Organiser who captured it).	Any (logged in)	None	200 OK – result object / 401 Unauthorized / 403 Forbidden / 404 Not Found

Routes (map/route info — a category can have zero or many routes)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories/{categoryId}/routes	Lists all route maps for a category.	None (public)	None	200 OK – list of routes / 404 Not Found
POST	/api/categories/{categoryId}/routes	Adds a route map to a category owned by the logged-in Organiser.	Organiser	{ mapFileUrl }	201 Created – route object / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
PUT	/api/routes/{id}	Updates a specific route map.	Organiser	{ mapFileUrl }	200 OK – updated route / 400 Bad Request / 401 Unauthorized / 403 Forbidden / 404 Not Found
DELETE	/api/routes/{id}	Deletes a specific route map.	Organiser	None	200 OK – confirmation / 401 Unauthorized / 403 Forbidden / 404 Not Found

Section C: SQL

```
SQLQuery1.sql...Student (52) X
1
2  -- RaceDay Database - Section C SQL Script
3
4  CREATE DATABASE RaceDayDB;
5
6  USE RaceDayDB;
7
8  -- CREATING TABLES
9  -- 1. ORGANISER TABLE
10 CREATE TABLE Organiser (
11     OrganiserId VARCHAR(20) NOT NULL PRIMARY KEY,
12     FullName VARCHAR(100) NOT NULL,
13     Email VARCHAR(100) NOT NULL UNIQUE,
14     PasswordHash VARCHAR(255) NOT NULL
15 );
16
17 -- 2. PARTICIPANT TABLE
18 CREATE TABLE Participant (
19     ParticipantId VARCHAR(20) NOT NULL PRIMARY KEY,
20     FullName VARCHAR(100) NOT NULL,
21     Email VARCHAR(100) NOT NULL UNIQUE,
22     PasswordHash VARCHAR(255) NOT NULL,
23     DateOfBirth DATE NOT NULL,
24     Gender VARCHAR(20) NULL
25 );
26
27 -- 3. EVENT TABLE
28 CREATE TABLE Event (
29     EventId INT IDENTITY(1,1) PRIMARY KEY,
30     OrganiserId VARCHAR(20) NOT NULL,
31     EventName VARCHAR(150) NOT NULL,
32     Description VARCHAR(500) NULL,
33     EventDate DATE NOT NULL,
34     Location VARCHAR(150) NOT NULL,
35     EventType VARCHAR(20) NOT NULL DEFAULT 'Run'
36     CHECK (EventType IN ('Run', 'Walk', 'Cycle')),
37     CONSTRAINT FK_Event_Organiser FOREIGN KEY (OrganiserId) REFERENCES Organiser(OrganiserId)
38 );
39
40 -- 4. CATEGORY TABLE
41 CREATE TABLE Category (
42     CategoryId INT IDENTITY(1,1) PRIMARY KEY,
43     EventId INT NOT NULL,
44     CategoryName VARCHAR(100) NOT NULL,
45     Distance DECIMAL(6,2) NOT NULL,
46     CONSTRAINT FK_Category_Event FOREIGN KEY (EventId) REFERENCES Event(EventId)
47 );
48
49 -- 5. ROUTE TABLE
50 CREATE TABLE Route (
51     RouteId INT IDENTITY(1,1) PRIMARY KEY,
52     CategoryId INT NOT NULL,
53     MapFileUrl VARCHAR(255) NOT NULL,
54     CONSTRAINT FK_Route_Category FOREIGN KEY (CategoryId) REFERENCES Category(CategoryId)
55 );
56
57 -- 6. ENROLMENT TABLE
58 CREATE TABLE Enrolment (
59     EnrolmentId INT IDENTITY(1,1) PRIMARY KEY,
60     ParticipantId VARCHAR(20) NOT NULL,
61     CategoryId INT NOT NULL,
62     Status VARCHAR(20) NOT NULL DEFAULT 'Pending'
63     CHECK (Status IN ('Pending', 'Confirmed', 'Cancelled')),
64     EnrolledTime DATETIME NOT NULL DEFAULT GETDATE(),
65     CONSTRAINT FK_Enrolment_Participant FOREIGN KEY (ParticipantId) REFERENCES Participant(ParticipantId),
66     CONSTRAINT FK_Enrolment_Category FOREIGN KEY (CategoryId) REFERENCES Category(CategoryId)
67 );
68
69 -- 7. RESULT TABLE
70 CREATE TABLE Result (
71     ResultId INT IDENTITY(1,1) PRIMARY KEY,
72     EnrolmentId INT NOT NULL UNIQUE,
73     CapturedBy VARCHAR(20) NOT NULL,
74     FinishTime TIME NULL,
75     Position INT NULL,
76     CONSTRAINT FK_Result_Enrolment FOREIGN KEY (EnrolmentId) REFERENCES Enrolment(EnrolmentId),
77     CONSTRAINT FK_Result_Organiser FOREIGN KEY (CapturedBy) REFERENCES Organiser(OrganiserId)
78 );
79
```



```
SQLQuery1.s...tudent (52)
80
81 -- SAMPLE DATA
82
83 INSERT INTO Organisier (OrganiserId, FullName, Email, PasswordHash)
84 VALUES ('ORG001', 'Thabo Nkosi', 'thabo.nkosi@raceday.co.za', 'HASHED_PASSWORD_1'),
85 ('ORG002', 'Lindiwe Dlamini', 'lindiwe.dlamini@raceday.co.za', 'HASHED_PASSWORD_2');
86
87 INSERT INTO Participant (ParticipantId, FullName, Email, PasswordHash, DateOfBirth, Gender)
88 VALUES ('PART001', 'Sarah van der Merwe', 'sarah.vdm@example.com', 'HASHED_PASSWORD_3', '1995-03-12', 'Female'),
89 ('PART002', 'Sipho Mokoena', 'sipho.mokoena@example.com', 'HASHED_PASSWORD_4', '1990-07-25', 'Male');
90
91 INSERT INTO Event (OrganiserId, EventName, Description, EventDate, Location, EventType)
92 VALUES ('ORG001', 'Johannesburg City Run', 'A scenic 10km and 21km run through the streets of Johannesburg.', '2026-11-15', 'Johannesburg, Gauteng', 'Run'),
93 ('ORG002', 'Cape Town Cycle Challenge', 'A road cycling event along the Cape Town coastline.', '2026-10-03', 'Cape Town, Western Cape', 'Cycle'),
94 ('ORG001', 'Durban Beachfront Walk', 'A family-friendly walk along the Durban beachfront.', '2026-09-20', 'Durban, KwaZulu-Natal', 'Walk');
95
96 INSERT INTO Category (EventId, CategoryName, Distance)
97 VALUES (1, '10km', 10.00), (1, '21km', 21.10), (2, '50km', 50.00),
98 (2, '100km', 100.00), (3, '5km', 5.00);
99
100
101 INSERT INTO Route (CategoryId, MapFileUrl)
102 VALUES (1, 'https://raceday.blob.core.windows.net/routes/joburg-10km-map.pdf'),
103 (3, 'https://raceday.blob.core.windows.net/routes/capetown-50km-map.pdf');
104
105 INSERT INTO Enrolment (ParticipantId, CategoryId, Status, EnrolledTime)
106 VALUES ('PART001', 1, 'Confirmed', '2026-09-01 08:30:00'),
107 ('PART002', 4, 'Confirmed', '2026-09-02 10:15:00'),
108 ('PART001', 5, 'Pending', '2026-09-03 14:00:00');
109
110 INSERT INTO Result (EnrolmentId, CapturedBy, FinishTime, Position)
111 VALUES (1, 'ORG001', '00:52:30', 15), (2, 'ORG002', '03:10:05', 8);
112
113 SELECT * FROM Organisier;
114 SELECT * FROM Participant;
115 SELECT * FROM Event;
116 SELECT * FROM Category;
117 SELECT * FROM Route;
118 SELECT * FROM Enrolment;
119 SELECT * FROM Result;
```

SQLQuery1.s...tudent (52)

```
108 ('PART001', 5, 'Pending', '2026-09-03 14:00:00');
109
110 INSERT INTO Result (EnrolmentId, CapturedBy, FinishTime, Position)
111 VALUES (1, 'ORG001', '00:52:30', 15), (2, 'ORG002', '03:10:05', 8);
112
113 SELECT * FROM Organisier;
114 SELECT * FROM Participant;
115 SELECT * FROM Event;
116 SELECT * FROM Category;
117 SELECT * FROM Route;
118 SELECT * FROM Enrolment;
119 SELECT * FROM Result;
```

115 % 7 0

Results Messages

OrganiserId	FullName	Email	PasswordHash	
1	ORG001	Thabo Nkosi	thabo.nkosi@raceday.co.za	HASHED_PASSWORD_1
2	ORG002	Lindiwe Dlamini	lindiwe.dlamini@raceday.co.za	HASHED_PASSWORD_2

ParticipantId	FullName	Email	PasswordHash	DateOfBirth	Gender	
1	PART001	Sarah van der Merwe	sarah.vdm@example.com	HASHED_PASSWORD_3	1995-03-12	Female
2	PART002	Sipho Mokoena	sipho.mokoena@example.com	HASHED_PASSWORD_4	1990-07-25	Male

EventId	OrganiserId	EventName	Description	EventDate	Location	EventType	
1	1	ORG001	Johannesburg City Run	A scenic 10km and 21km run through the streets o...	2026-11-15	Johannesburg, Gauteng	Run
2	2	ORG002	Cape Town Cycle Challenge	A road cycling event along the Cape Town coastli...	2026-10-03	Cape Town, Western Cape	Cycle
3	3	ORG001	Durban Beachfront Walk	A family-friendly walk along the Durban beachfront.	2026-09-20	Durban, KwaZulu-Natal	Walk

CategoryId	EventId	CategoryName	Distance
1	1	10km	10.00
2	2	21km	21.10
3	3	50km	50.00
4	4	100km	100.00
5	5	5km	5.00

RouteId	CategoryId	MapFileUrl
1	1	https://raceday.blob.core.windows.net/routes/job...
2	3	https://raceday.blob.core.windows.net/routes/cap...

EnrolmentId	ParticipantId	CategoryId	Status	EnrolledTime	
1	1	PART001	1	Confirmed	2026-09-01 08:30:00.000
2	2	PART002	4	Confirmed	2026-09-02 10:15:00.000
3	3	PART001	5	Pending	2026-09-03 14:00:00.000

ResultId	EnrolmentId	CapturedBy	FinishTime	Position	
1	1	1	ORG001	00:52:30.0000000	15
2	2	2	ORG002	03:10:05.0000000	8

Query executed successfully.

References

Coronel, Carlos; Morris, Steven; Crockett, Keeley; Blewett, Craig: Database Principles, fundamentals and design, implementation, and management 3rd Edition, 2020

Cloude:

<https://www.bing.com/ck/a?!&p=e5b8847f2ad0e38fff39dedd1528be738e943bf530038a1556502a529f08d31dJmItldHM9MTc4ODQ4MDAwMA&ptn=3&ver=2&hsh=4&fclid=321ac19a-0303-6a17-133b-d65002406b81&psq=cloude+ai&u=a1aHR0cHM6Ly9jbGF1ZGUuY29tLw>